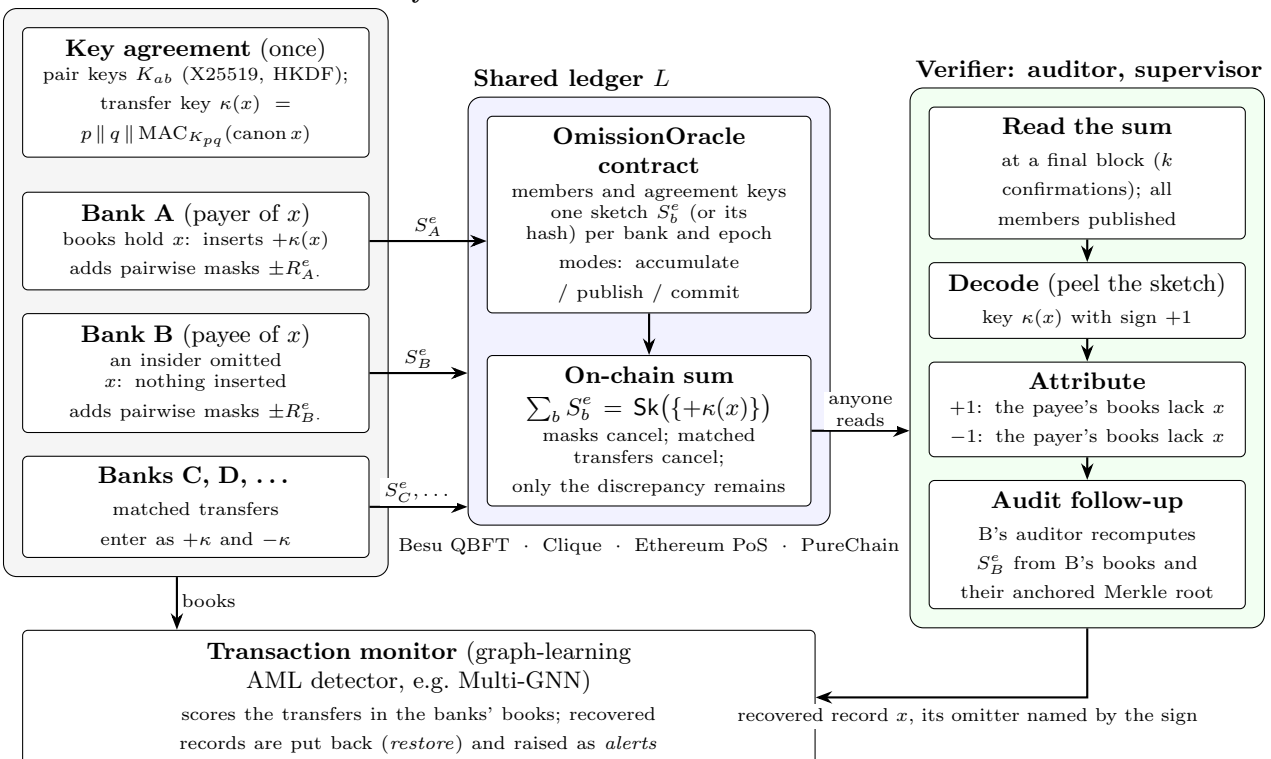

Consortium members: books stay off chain
every epoch e (e.g. hourly)
Key agreement (once)
pair keys K_{ab} (X25519, HKDF);
transfer key $\kappa(x)$ =
$p \,\|\, q \,\|\, \mathrm{MAC}_{K_{pq}}(\mathrm{canon}\, x)$
Bank A (payer of x)
books hold x: inserts $+\kappa(x)$
adds pairwise masks $\pm R^e_{A\cdot}$
Bank B (payee of x)
an insider omitted
x: nothing inserted
adds pairwise masks $\pm R^e_{B\cdot}$
Banks C, D, ...
matched transfers
enter as $+\kappa$ and $-\kappa$
S^e_A
S^e_B
$S^e_C, \ldots$
Shared ledger L
OmissionOracle contract
members and agreement keys
one sketch S^e_b (or its hash) per bank and epoch
modes: accumulate / publish / commit
On-chain sum
$\sum_b S^e_b = \mathsf{Sk}(\{+\kappa(x)\})$
masks cancel; matched transfers cancel;
only the discrepancy remains
Besu QBFT · Clique · Ethereum PoS · PureChain
anyone reads
Verifier: auditor, supervisor
Read the sum
at a final block (k confirmations); all members published
Decode (peel the sketch)
key $\kappa(x)$ with sign $+1$
Attribute
$+1$: the payee's books lack x
-1: the payer's books lack x
Audit follow-up
B's auditor recomputes S^e_B from B's books and their anchored Merkle root
books
Transaction monitor (graph-learning AML detector, e.g. Multi-GNN)
scores the transfers in the banks' books; recovered records are put back (restore) and raised as alerts
recovered record x, its omitter named by the sign